

TOPOLOGICAL GAMES, RAMSEY THEORY, AND METRIC NUMBER THEORY

PRELIMINARY NOTES PREPARED BY J. IOVINO

1. THE ELLENTUCK SPACE AND COMPLETELY RAMSEY SETS

Definition 1.1. Let $\mathbb{N} = \{0, 1, 2, \dots\}$ be the set of non-negative integers. We denote by $[\mathbb{N}]^\omega$ the collection of all infinite subsets of $\mathbb{N}$, and by $[\mathbb{N}]^{<\omega}$ the collection of all finite subsets of $\mathbb{N}$. For $A \in [\mathbb{N}]^\omega$, we write $[A]^\omega$ for the set of all infinite subsets of A . For $s \in [\mathbb{N}]^{<\omega}$ and $A \in [\mathbb{N}]^\omega$, we write $s < A$ to mean that either $s = \emptyset$ or $\max(s) < \min(A)$. If $s < A$, the *combinatorial interval* $[s, A]$ is defined by

$$[s, A] = \{X \in [\mathbb{N}]^\omega : s \subseteq X \subseteq s \cup A \text{ and } s = X \cap \{0, 1, \dots, \max(s)\}\}.$$

The *Ellentuck space* is the set $[\mathbb{N}]^\omega$ equipped with the topology τ_E generated by the base of all such intervals $[s, A]$.

Definition 1.2. A set $\mathcal{X} \subseteq [\mathbb{N}]^\omega$ is said to be *completely Ramsey* if for every $s \in [\mathbb{N}]^{<\omega}$ and every $A \in [\mathbb{N}]^\omega$ with $s < A$, there exists $B \in [A]^\omega$ such that either

$$[s, B] \subseteq \mathcal{X} \quad \text{or} \quad [s, B] \cap \mathcal{X} = \emptyset.$$

When $s = \emptyset$, this condition asserts the existence of an infinite set $B \subseteq A$ such that $[B]^\omega \subseteq \mathcal{X}$ or $[B]^\omega \cap \mathcal{X} = \emptyset$, which is the partition property of the infinite Ramsey theorem.

Definition 1.3. Let (X, τ) be a topological space. A set $E \subseteq X$ is said to have the *property of Baire* in (X, τ) if there exists an open set $U \in \tau$ such that the symmetric difference $E \triangle U = (E \setminus U) \cup (U \setminus E)$ is of the first category in (X, τ) .

Theorem 1.4 (Ellentuck). Let τ_E be the Ellentuck topology on $[\mathbb{N}]^\omega$.

- (1) A set $\mathcal{N} \subseteq [\mathbb{N}]^\omega$ is nowhere dense in $([\mathbb{N}]^\omega, \tau_E)$ if and only if for every interval $[s, A]$, there exists $B \in [A]^\omega$ such that $[s, B] \cap \mathcal{N} = \emptyset$.
- (2) Every set of the first category in $([\mathbb{N}]^\omega, \tau_E)$ is nowhere dense.
- (3) A set $\mathcal{X} \subseteq [\mathbb{N}]^\omega$ is completely Ramsey if and only if $\mathcal{X}$ has the property of Baire in $([\mathbb{N}]^\omega, \tau_E)$.

Definition 1.5. Let $\mathcal{X} \subseteq [\mathbb{N}]^\omega$, $s \in [\mathbb{N}]^{<\omega}$, and $A \in [\mathbb{N}]^\omega$ with $s < A$. The *Kastanas game* on $[s, A]$ with target $\mathcal{X}$, denoted by $G_K(\mathcal{X}, [s, A])$, is played as follows.

- (1) In round 0, Player I chooses $A_0 \in [A]^\omega$. Player II responds with an element $a_0 \in A_0$ and a set $B_0 \in [A_0]^\omega$ with $a_0 < B_0$.
- (2) In round $n \geq 1$, Player I chooses $A_n \in [B_{n-1}]^\omega$. Player II responds with $a_n \in A_n$ and $B_n \in [A_n]^\omega$ with $a_n < B_n$.

Since $a_n \in A_n \subseteq B_{n-1}$ and $a_{n-1} < B_{n-1}$, the play yields a sequence $a_0 < a_1 < a_2 < \dots$, and hence the set

$$X = s \cup \{a_0, a_1, a_2, \dots\} \in [s, A].$$

Player II wins the play if $X \in \mathcal{X}$; otherwise, Player I wins.



Remark 1.6. After round n the play is confined to the basic open set $[s \cup \{a_0, \dots, a_n\}, B_n]$, and these sets decrease. Thus the Kastanas game is a Banach–Mazur-type game on the Ellentuck space, in which the players shrink basic open sets. Note that Player II must be allowed to shrink the set as well as choose a point: if she could only choose points, Player I could confine the play to any infinite subset of A he liked.

Theorem 1.7 (Kastanas). *Let $\mathcal{X} \subseteq [\mathbb{N}]^\omega$, $s \in [\mathbb{N}]^{<\omega}$, and $A \in [\mathbb{N}]^\omega$ with $s < A$.*

- (1) *Player I has a winning strategy in $G_K(\mathcal{X}, [s, A])$ if and only if there exists $B \in [A]^\omega$ such that $[s, B] \cap \mathcal{X} = \emptyset$.*
- (2) *Player II has a winning strategy in $G_K(\mathcal{X}, [s, A])$ if and only if for every $A' \in [A]^\omega$ there exists $B \in [A']^\omega$ such that $[s, B] \subseteq \mathcal{X}$.*

Consequently, $\mathcal{X}$ is completely Ramsey if and only if the game $G_K(\mathcal{X}, [s, A])$ is determined for every interval $[s, A]$.

Proof. We prove the “if” directions and the consequence; the “only if” directions are proved by a fusion argument that builds the set B from a winning strategy, for which we refer to [Kas83].

For (1), suppose $[s, B] \cap \mathcal{X} = \emptyset$ with $B \in [A]^\omega$. Player I plays $A_0 = B$ and then $A_n = B_{n-1}$. Every a_n then lies in B , so $X \in [s, B]$ and $X \notin \mathcal{X}$.

For (2), suppose the condition holds. In round 0, Player II chooses $B \in [A_0]^\omega$ with $[s, B] \subseteq \mathcal{X}$ and plays $a_0 = \min B$ and $B_0 = B \setminus \{a_0\}$. In round $n \geq 1$ she plays $a_n = \min A_n$ and $B_n = A_n \setminus \{a_n\}$. Since $A_n \subseteq B_{n-1} \subseteq B$ for all n , every a_n lies in B , so $X \in [s, B] \subseteq \mathcal{X}$.

For the consequence, suppose first that $\mathcal{X}$ is completely Ramsey, and fix $[s, A]$. If some $B \in [A]^\omega$ has $[s, B] \cap \mathcal{X} = \emptyset$, Player I wins by (1). Otherwise, for each $A' \in [A]^\omega$ the Ramsey property applied to $[s, A']$ gives $B \in [A']^\omega$ with $[s, B] \subseteq \mathcal{X}$, and Player II wins by (2). Conversely, if every such game is determined, then for each $[s, A]$ either (1) gives $B \in [A]^\omega$ with $[s, B] \cap \mathcal{X} = \emptyset$, or (2), applied with $A' = A$, gives $B \in [A]^\omega$ with $[s, B] \subseteq \mathcal{X}$. 😊

Theorem 1.8. *For $\mathcal{X} \subseteq [\mathbb{N}]^\omega$, the following statements are equivalent:*

- (1) *$\mathcal{X}$ is completely Ramsey.*
- (2) *$\mathcal{X}$ has the property of Baire in the Ellentuck topology τ_E .*
- (3) *The Kastanas game $G_K(\mathcal{X}, [s, A])$ is determined for every interval $[s, A]$.*

Proof. The equivalence of (1) and (2) is Theorem 1.4(3), and the equivalence of (1) and (3) is Theorem 1.7. 😊

Corollary 1.9 (Galvin–Prikry). *Every Borel subset of $[\mathbb{N}]^\omega$ (with respect to the standard metric topology) is completely Ramsey.*

Proof. The standard metric topology on $[\mathbb{N}]^\omega$ is the subspace topology inherited from $\{0, 1\}^\mathbb{N}$, which is coarser than the Ellentuck topology τ_E . Hence every Borel set in the metric topology is Borel in $([\mathbb{N}]^\omega, \tau_E)$. In any topological space the sets with the property of Baire form a σ -algebra containing the open sets, so every Borel set $\mathcal{X}$ has the property of Baire in τ_E , and is completely Ramsey by Theorem 1.4(3). Alternatively, the payoff set of $G_K(\mathcal{X}, [s, A])$ is the preimage of $\mathcal{X}$ under a continuous map from the space of plays, hence Borel, so the game is determined by Martin’s Borel determinacy theorem [Mar75], and Theorem 1.7 applies. 😊

2. SCHMIDT GAMES AND BADLY APPROXIMABLE NUMBERS

Definition 2.1. In the metric theory of Diophantine approximation, a real number $x \in \mathbb{R}$ is said to be *badly approximable* if there exists a constant $c > 0$ such that

$$\left| x - \frac{p}{q} \right| \geq \frac{c}{q^2}$$

for all $p \in \mathbb{Z}$ and all $q \in \mathbb{N} \setminus \{0\}$. The collection of all such numbers is denoted by **Bad**.



Remark 2.2. The set **Bad** is of Lebesgue measure zero by Khintchine's theorem, and is of the first category in $\mathbb{R}$. Consequently, neither measure theory nor the classical Banach–Mazur game can distinguish **Bad** from a negligible set. To study such sets, W. M. Schmidt (1966) introduced a metric modification of the Banach–Mazur game.

Definition 2.3. Let (X, d) be a complete metric space, and let $0 < \alpha < 1$ and $0 < \beta < 1$. For any closed ball $B = B(c, r) = \{x \in X : d(x, c) \leq r\}$, we denote its radius by $\rho(B) = r$. The (α, β) -Schmidt game on X with target set $S \subseteq X$ is played as follows.

Player I chooses a closed ball $B_0 \subseteq X$. Player II chooses a closed ball $W_0 \subseteq B_0$ with $\rho(W_0) = \alpha\rho(B_0)$. At step $n \geq 1$, Player I chooses a closed ball $B_n \subseteq W_{n-1}$ with $\rho(B_n) = \beta\rho(W_{n-1})$, and Player II responds with a closed ball $W_n \subseteq B_n$ with $\rho(W_n) = \alpha\rho(B_n)$.

Since X is complete and $\rho(B_n) \rightarrow 0$, the Cantor intersection theorem implies that

$$\bigcap_{n=0}^{\infty} B_n = \bigcap_{n=0}^{\infty} W_n = \{x\}$$

for a unique point $x \in X$. Player II *wins* the play if $x \in S$. The set S is called (α, β) -*winning* if Player II has a winning strategy. A set S is called α -*winning* if it is (α, β) -winning for all $\beta \in (0, 1)$, and *winning* if it is α -winning for some $\alpha \in (0, 1)$.

Theorem 2.4 (Schmidt). *Let $S \subseteq \mathbb{R}^n$ be a winning set.*

- (1) *The set S is dense in $\mathbb{R}^n$.*
- (2) *The Hausdorff dimension of S is maximal: $\dim_H(S) = n$.*
- (3) *If $\{S_k\}_{k=1}^{\infty}$ is a countable sequence of α -winning sets, then $\bigcap_{k=1}^{\infty} S_k$ is an α -winning set, and hence has Hausdorff dimension n .*

Theorem 2.5 (Schmidt). *The set **Bad** of badly approximable numbers is α -winning in $\mathbb{R}$ for every $0 < \alpha < \frac{1}{2}$. In particular, **Bad** has Hausdorff dimension 1. Moreover, if $f_1, f_2, \dots$ are similarities $f_i(x) = c_i x + d_i$ of $\mathbb{R}$ ($c_i \neq 0$), then $\bigcap_i f_i(\mathbf{Bad})$ has Hausdorff dimension 1; for example, every real number is the sum of two badly approximable numbers.*

Proof. Fix $0 < \alpha < \frac{1}{2}$ and $0 < \beta < 1$. Write $\rho_k = \rho(B_k)$, so that $\rho_k = (\alpha\beta)^k \rho_0$, and put $Q_k = (8\rho_k)^{-1/2}$ and $Q_{-1} = 0$. Two distinct rationals p/q and p'/q' with $q, q' \leq Q_k$ satisfy

$$\left| \frac{p}{q} - \frac{p'}{q'} \right| \geq \frac{1}{qq'} \geq \frac{1}{Q_k^2} = 8\rho_k,$$

so at most one of them lies within distance ρ_k of B_k , a set contained in an interval of length $4\rho_k$.

Player II plays as follows. In answer to B_k , consider the two intervals of radius $\alpha\rho_k$ at the two ends of B_k . Since $\alpha < \frac{1}{2}$, they are separated by a gap of length $2(1 - 2\alpha)\rho_k$, so every point is at distance at least $(1 - 2\alpha)\rho_k$ from one of them. If some rational p/q with $Q_{k-1} < q \leq Q_k$ lies within distance ρ_k of B_k (there is at most one), Player II lets W_k be the end interval farther from p/q ; otherwise she plays either one.

Let x be the point determined by a play consistent with this strategy, and let p/q be any rational. Choose k with $Q_{k-1} < q \leq Q_k$. Since $x \in W_k \subseteq B_k$, the rule for round k gives $|x - p/q| \geq (1 - 2\alpha)\rho_k$. If $k \geq 1$, then $q > Q_{k-1}$ gives $1/q^2 < 8\rho_{k-1} = 8\rho_k/(\alpha\beta)$; if $k = 0$, then $1/q^2 \leq 1$. In either case $|x - p/q| \geq c/q^2$ with $c = (1 - 2\alpha)\min\{\alpha\beta/8, \rho_0\}$. Hence $x \in \mathbf{Bad}$, and $\mathbf{Bad}$ is (α, β) -winning for every β .

The dimension statement follows from Theorem 2.4(2). A similarity maps balls to balls and multiplies all radii by the same factor, so it maps α -winning sets to α -winning sets; the statement about $\bigcap_i f_i(\mathbf{Bad})$ then follows from Theorem 2.4(2),(3). For the last assertion, given $x \in \mathbb{R}$, the set $\mathbf{Bad} \cap (x - \mathbf{Bad})$ is non-empty; if b lies in it, then $x = b + (x - b)$ with $b, x - b \in \mathbf{Bad}$. 



Remark 2.6 (Homogeneous Dynamics and the Dani Correspondence). The connection between Schmidt games and modern number theory extends into ergodic theory and Lie groups through the *Dani correspondence* (1985). Let $G = \mathrm{SL}(m+n, \mathbb{R})$ and $\Gamma = \mathrm{SL}(m+n, \mathbb{Z})$. The quotient space $X = G/\Gamma$ is the space of unimodular lattices in $\mathbb{R}^{m+n}$. For an $m \times n$ matrix A , define the one-parameter diagonal flow $g_t = \mathrm{diag}(e^{t/m}I_m, e^{-t/n}I_n)$ on X , and let

$$u_A = \begin{pmatrix} I_m & A \\ 0 & I_n \end{pmatrix}.$$

Dani proved that A is a badly approximable system of linear forms if and only if the forward orbit $\{g_t u_A \Gamma : t \geq 0\}$ is *bounded* in X .

Schmidt's game enters this picture through the work of Dani, who used it to show that the points with bounded orbits under certain such flows form a winning set; see Section 3. McMullen introduced the *absolute* variant of Schmidt's game [McM10], which is preserved by quasiconformal maps.

3. TOPOLOGICAL GAMES AND OPEN PROBLEMS IN CLASSICAL MATHEMATICS

The games of these notes are more than a reformulation of known facts. In the examples below, a game was either the decisive tool in solving a long-standing problem, or gave the strongest form of a result first proved by other means.

Banach's homogeneous space problem. In his 1932 monograph, Banach asked whether a Banach space that is isomorphic to each of its infinite-dimensional closed subspaces must be isomorphic to the Hilbert space ℓ_2 . The answer, yes, was obtained in the mid-1990s by combining two results. Komorowski and Tomczak-Jaegermann [KTJ95] showed that a homogeneous space not isomorphic to ℓ_2 would contain a subspace with no unconditional basis. Gowers [Gow96] proved his *dichotomy theorem*: every infinite-dimensional Banach space contains either a subspace with an unconditional basis or a hereditarily indecomposable subspace, and a hereditarily indecomposable space is not isomorphic to any of its proper subspaces. The heart of Gowers's proof is an infinite-dimensional Ramsey theorem for block sequences, whose conclusion is stated in terms of a two-player game: one player proposes block subspaces, the other picks vectors from them, and the theorem says that, after passing to a suitable subspace, one of the two alternatives of the dichotomy is witnessed by a winning strategy [Gow02]. It plays the same role for Banach spaces that the Banach–Mazur game on the Ellentuck space plays for the Galvin–Prikry and Ellentuck theorems of Section 1. The work was cited in Gowers's 1998 Fields Medal.

Schmidt’s conjecture. For $s, t \geq 0$ with $s + t = 1$, let $\mathbf{Bad}(s, t)$ be the set of $(x, y) \in \mathbb{R}^2$ for which there is $c > 0$ with

$$\max\{\|qx\|^{1/s}, \|qy\|^{1/t}\} \geq \frac{c}{q} \quad \text{for all } q \in \mathbb{N} \setminus \{0\},$$

where $\|\cdot\|$ denotes the distance to the nearest integer. In 1983 Schmidt conjectured that $\mathbf{Bad}(\frac{1}{3}, \frac{2}{3}) \cap \mathbf{Bad}(\frac{2}{3}, \frac{1}{3}) \neq \emptyset$, and observed that a counterexample would imply the Littlewood conjecture. Badziahin, Pollington, and Velani [BPV11] proved the conjecture, and in fact that a countable intersection $\bigcap_i \mathbf{Bad}(s_i, t_i)$ has full Hausdorff dimension provided $\liminf_i \min\{s_i, t_i\} > 0$. Their proof does not use games. An [An16] then proved that each $\mathbf{Bad}(s, t)$ is winning for Schmidt’s game. Since countable intersections of α -winning sets are α -winning (Theorem 2.4), this gives a short proof of Schmidt’s conjecture and removes the $\liminf$ condition. Beresnevich, Nesharim, and Yang [BNY21] extended this to all dimensions: for every weight vector $\mathbf{w}$, the set $\mathbf{Bad}(\mathbf{w}) \subseteq \mathbb{R}^n$ is *hyperplane absolute winning*, a variant of Schmidt’s game that is also stable under countable intersections.

Bounded orbits in homogeneous dynamics. By the Dani correspondence (Remark 2.6), badly approximable systems of linear forms correspond to bounded orbits of a diagonal flow on the space of lattices. Dani [Dan86] used Schmidt’s game to show that, for certain flows on such spaces, the points with bounded orbits form a winning set, hence a dense set of full Hausdorff dimension. Margulis conjectured that the same abundance holds for every non-quasiunipotent flow on a finite-volume homogeneous space. Kleinbock and Margulis [KM96] proved the full-dimension statement in general, by methods based on mixing rather than games; whether the sets of bounded orbits are winning in this generality has since been established only in special cases.

4. LATTICES BEYOND NUMBER THEORY: COMMON GROUND AND OPEN QUESTIONS

The space of lattices that appears in Remark 2.6 also underlies two active areas outside pure mathematics. We record the common ground, and state plainly that no connection between topological games and either area is known.

Definition 4.1. A *lattice* of rank n in $\mathbb{R}^m$ is a subgroup $\Lambda = \{\sum_{i=1}^n x_i b_i : x_i \in \mathbb{Z}\}$ generated by linearly independent vectors $b_1, \dots, b_n$. Its *first minimum* is $\lambda_1(\Lambda) = \min\{\|v\| : v \in \Lambda \setminus \{0\}\}$. Let $X_n = \mathrm{SL}(n, \mathbb{R})/\mathrm{SL}(n, \mathbb{Z})$ be the space of unimodular lattices in $\mathbb{R}^n$. *Mahler’s compactness criterion* states that a subset $K \subseteq X_n$ is relatively compact if and only if there is $\epsilon > 0$ with $\lambda_1(\Lambda) \geq \epsilon$ for all $\Lambda \in K$.



Remark 4.2 (Lattice-based cryptography). The *shortest vector problem* SVP_γ asks, given a basis of a lattice Λ , for a non-zero $v \in \Lambda$ with $\|v\| \leq \gamma \lambda_1(\Lambda)$. In the *learning with errors* problem LWE one must distinguish pairs $(a_i, \langle a_i, s \rangle + e_i)$ in $\mathbb{Z}_q^n \times \mathbb{Z}_q$, with a secret s and small random errors e_i , from uniformly random pairs. Ajtai [Ajt96] gave the first reduction from worst-case lattice problems to an average-case problem, and Regev [Reg05] showed that solving LWE on average is at least as hard as solving approximate worst-case lattice problems, with approximation factors polynomial in n , for *quantum* algorithms; a classical reduction, for a variant of SVP , was later given by Peikert [Pei09]. These results concern lattices of large dimension n and the running time of algorithms. The Schmidt-game results of Section 3 concern Lebesgue-null sets of parameters and yield Hausdorff dimension; they say nothing about algorithmic hardness.



Remark 4.3 (Quantum error correction). Consider a Gottesman–Kitaev–Preskill code [GKP01] on n bosonic modes. Its stabilizers are displacement operators indexed by a lattice

$\Lambda \subseteq \mathbb{R}^{2n}$ that is contained in its symplectic dual $\Lambda^\perp$. Logical errors correspond to displacements in $\Lambda^\perp \setminus \Lambda$, so the length of a shortest such vector plays the role of the code distance, and decoding under shift noise is a closest-vector problem. The design of good codes is thus a question about packing and covering properties of symplectic lattices; see [CES22] for a survey from this point of view. Again, these are questions about specific lattices and their optimality, on which winning-set results give no information.



Remark 4.4. Whether game-theoretic methods have anything to contribute to these areas is, as far as we know, an open question.

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